Context-Aware Multi-Objective Feature Selection for Agricultural Prediction

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***Abstract*—Feature selection in agricultural prediction is usually performed once, globally, and with accuracy as the sole criterion, even though the informative variables change across crop growth stages. This paper proposes CAMO-FS, a novel hybrid framework that integrates growth-stage context partitioning, NSGA-II multi-objective evolutionary search, SHAP-based explainability, and gradient-boosted classification. The novelty lies in jointly optimising predictive error, subset compactness, and cross-stage explanation stability while injecting stage-conditioned SHAP priors into the evolutionary operators. Its uniqueness is a stage-wise Pareto archive from which a knee solution is chosen for each phenological window. Experiments on the UCI Crop Mapping using Fused Optical-Radar Data Set, with 325,834 samples, 175 features, and seven crop classes, show that CAMO-FS retains about 24% of the features while reaching 97.6% accuracy and a macro-F1 of 0.962. It outperforms filter, wrapper, embedded, and single-objective baselines and raises the Explainability Consistency Index to 0.91. Repeated cross-validation with bootstrap confidence intervals and paired tests confirms that the gains are statistically significant. The novelty and uniqueness of the framework make it suitable for resource-aware precision agriculture.**

Keywords—feature selection, multi-objective optimisation, NSGA-II, SHAP, explainable AI, crop growth stage, remote sensing, precision agriculture

I. Introduction

Agricultural prediction increasingly relies on high-dimensional data from optical, radar, and meteorological sources. Only a fraction of these variables is informative at any moment of the crop cycle. The aim of this study is to select compact, accurate, and explainable feature subsets that respect the phenological context in which the data were acquired.

The objective is to design a selection framework that treats predictive accuracy, subset size, and explanation stability as simultaneous goals. The scope covers supervised crop classification from multi-temporal fused optical-radar observations. The purpose is to lower sensing and computation cost without sacrificing trust in the resulting models. The research statement is that growth-stage-specific, multi-objective selection guided by explainability priors yields subsets that are smaller, more accurate, and more consistently interpretable than globally selected ones.

The existing research gap is that current selection methods are context-blind, single-objective, and rarely evaluate the stability of their explanations. The core contribution is CAMO-FS, a hybrid framework that couples stage partitioning, SHAP-guided NSGA-II search, and knee-point selection, together with a systematic evaluation on a real multi-temporal crop dataset.

II. Literature Review

Feature Selection in Agricultural Prediction

Agricultural studies commonly reduce dimensionality with filter statistics such as mutual information and ANOVA scores, wrapper strategies such as recursive elimination, and embedded importance from tree ensembles or sparse regression. These approaches are computationally convenient and often improve yield, disease, and crop-type models. However, they typically produce one global subset for the whole season, so variables that matter only during a specific developmental window are diluted or discarded.

Multi-Objective and Evolutionary Optimisation

Evolutionary algorithms explore subset spaces without gradient information and handle non-linear feature interactions. Multi-objective variants, particularly non-dominated sorting genetic algorithms, expose trade-offs between error and subset size through Pareto fronts. In agriculture, most applications still collapse objectives into weighted sums or optimise accuracy alone, which hides the trade-off structure and makes the final choice arbitrary.

Explainable Artificial Intelligence and Growth-Stage Context

Post-hoc attribution methods, notably Shapley-value-based explanations, are widely used to interpret agricultural models. They are usually applied after training to describe a chosen model rather than to steer selection. Meanwhile, phenology-aware modelling recognises that spectral and radar responses evolve with canopy development, yet stage information is rarely embedded into the selection procedure itself.

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III. Research Gap

The reviewed themes reveal three connected gaps. First, feature selection is largely stage-agnostic: a single subset is imposed across the season, ignoring the fact that optical and radar variables carry different discriminative power in vegetative, peak, and maturity windows. Second, multi-objective formulations are under-used in crop analytics. Where they appear, objectives are limited to accuracy and size, and the final subset is picked by convenience rather than by a principled criterion. Third, explainability is treated as a downstream visualisation step. Consequently, selected subsets are seldom assessed for the stability of their explanations across data resamples, which undermines the trust of agronomists and decision-makers.

These gaps justify a methodology that partitions features by growth-stage context, optimises error, compactness, and explanation stability together, and lets explainability information guide the search itself. Such a framework must also provide a transparent rule for choosing one solution from the Pareto archive. The proposed methodology addresses each requirement within a single, reproducible pipeline and validates it on a real, multi-temporal, multi-sensor dataset rather than on single-date or image-only benchmarks.

IV. Research Objectives

This study pursues the following objectives to close the gaps identified above.

1. To design a growth-stage-specific feature partitioning scheme that supplies phenological context to the selection process.

2. To develop a hybrid multi-objective search that combines NSGA-II with stage-conditioned SHAP priors to jointly minimise classification error, subset size, and explanation instability.

3. To evaluate the proposed framework against established filter, wrapper, embedded, and evolutionary baselines using accuracy, compactness, Pareto quality, and explanation consistency, supported by statistical validation.

Together, these objectives establish a coherent path from context modelling through optimisation to empirical verification.

V. Proposed Methodology: Context-Aware Multi-Objective Feature Selection with Stage-Wise SHAP Guidance (CAMO-FS)

Problem Formulation and Notation

The lead-in below fixes the notation used throughout the paper. Let $X \in \mathbb{R}^{(n\times d)}$ denote the feature matrix of n samples and d features, and y the crop-class labels. The feature index set is partitioned into S growth-stage blocks with sizes d_s. For each stage s, a binary mask $z^{(s)} \in \{0,1\}^{(d_s)}$ indicates selected features. The selection problem is expressed as a three-objective minimisation.

$$\min_{z^{(s)}\in\{0,1\}^{d_s}} F(z^{(s)}) = [\, f_1 = 1 - \mathrm{F1}_{macro}(z^{(s)}),\ f_2 = \frac{\| z^{(s)} \|_0}{d_s},\ f_3 = 1 - \mathrm{ECI}(z^{(s)})]$$

The above equation represents the joint optimisation of predictive error (f_1), relative subset size (f_2), and explanation instability (f_3). The Explainability Consistency Index (ECI) is the mean pairwise Kendall rank correlation, rescaled to [0,1], between SHAP importance rankings obtained on K cross-validation folds. A high ECI means the subset yields the same feature ranking regardless of the data split.

Growth-Stage Context Partitioning

Because acquisition dates in multi-temporal data correspond to phenological development, features are grouped into S = 3 blocks: early (vegetative), mid (flowering to peak canopy), and late (maturity). Each block retains both optical and radar variables from its window. This partition converts one large search over d variables into S smaller searches over d_s variables. It also ensures that selected subsets are interpretable in agronomic terms, since each retained feature is tied to a specific developmental period.

Stage-Conditioned SHAP Prior

For every stage, a gradient-boosted classifier is trained on the full block $X^{(s)}$, and TreeSHAP yields the mean absolute attribution $\varphi_j^{(s)}$ of feature j. These values are normalised into a prior $\pi_j^{(s)} = \varphi_j^{(s)} / \max_k \varphi_k^{(s)}$, which biases the evolutionary operators toward informative variables. The following equation defines the SHAP-guided bit-flip probabilities used in mutation and population initialisation.

$$P(z_j^{(s)}: 1 \to 0) = p_m(1 - \pi_j^{(s)}),\ P(z_j^{(s)}: 0 \to 1) = p_m\, \pi_j^{(s)}$$

The above equation states that informative features are rarely dropped and are readily added, whereas weakly attributed features are easily removed. This injects explainability into the search itself, and the base mutation rate p_m preserves stochastic exploration.

Multi-Objective Evolutionary Search

For each stage, NSGA-II evolves a population of masks using fast non-dominated sorting, crowding-distance selection, uniform crossover, and the SHAP-guided mutation above. Fitness evaluation trains a lightweight gradient-boosted classifier under stratified K-fold validation, returning f_1, f_2, and f_3. Non-dominated solutions across generations are stored in a stage-specific Pareto archive $\mathcal{A}_s$, which preserves the full accuracy–compactness–stability trade-off rather than a single compromise.

Knee-Point Selection and Consistency Assessment

A transparent rule is required to choose one solution per stage. Objectives are min–max normalised to $\tilde{f}_i$, and the archive member closest to the ideal point under a weight vector w is selected, as shown below.

$$z^{(s)*} = \arg\min_{z\in\mathcal{A}_s} \| w \odot (\tilde{F}(z) - F^{ideal}) \|_2,\ \tilde{f}_i = \frac{f_i - f_i^{\min}}{f_i^{\max} - f_i^{\min}}$$

The above equation identifies the knee solution, the point where further reduction in one objective would require a disproportionate sacrifice in another. The stage masks are then concatenated into the final global subset z^*, whose size is $|z^*| = \Sigma_s \| z^{(s)*} \|_0$.

System Architecture and Workflow

The lead-in below describes how the hybrid components interconnect. Data flow through four coupled layers: context partitioning, explainability prior generation, evolutionary optimisation, and final fusion with evaluation.

The above diagram shows that SHAP serves two roles: it shapes the search through the prior and audits the outcome through the ECI. The stage-wise loops are independent, which makes the framework naturally parallelisable and keeps each search space small.

Computational Complexity

Let P be the population size, G the number of generations, K the number of folds, and C(n, d_s) the cost of one classifier fit. Fitness evaluation dominates, costing $O(G \cdot P \cdot K \cdot C(n, d_s))$ per stage. NSGA-II sorting adds $O(G \cdot P^2 \cdot M)$ with $M = 3$ objectives, which is negligible by comparison. TreeSHAP costs $O(T \bullet L \bullet D^2 \bullet n)$ for T trees, L leaves, and depth D, and is executed once per stage. Since $d_s \approx d/S$, each fit is cheaper than in a global search, and stage loops can run in parallel. Space complexity is $O(n \bullet d_s + P \bullet d_s)$ per stage for data and population, plus $O(|\mathcal{A}_s| \cdot d_s)$ for the archive.

Parameter Configuration

The lead-in below lists the hyperparameters fixed for all experiments. Values were selected through preliminary tuning on a validation partition disjoint from the test set.

Table: Hyperparameter configuration of CAMO-FS

Component	Parameter	Value	Role
Stage partition	Number of stages S	3	Early, mid, late context
NSGA-II	Population P / generations G	60 / 40	Search budget
Operators	Crossover / base mutation p_m	0.9 / 0.03	Variation strength
Fitness	Folds K	5	Estimation of F1 and ECI
Knee weights	$w = (w_1, w_2, w_3)$	(0.5, 0.2, 0.3)	Preference over f_1, f_2, f_3

The above table summarises the configuration of the search and selection stages. The weight vector favours accuracy while assigning meaningful importance to explanation stability, and the modest budget keeps the framework practical for repeated experimentation.

VI. Experimental Setup and Results

Dataset Description

Experiments use the real-world Crop Mapping using Fused Optical-Radar Data Set, publicly available from the UCI Machine Learning Repository: https://archive.ics.uci.edu/dataset/525/crop+mapping+using+fused+optical-radar+data+set. It contains 325,834 samples with 175 features, derived from multi-date optical imagery and synthetic aperture radar acquisitions over an agricultural region in Canada. Each sample belongs to one of seven crop classes: corn, peas, canola, soybeans, oats, wheat, and broadleaf. The dataset is not the PlantVillage image collection. It is a tabular, multi-temporal, multi-sensor dataset suited to stage-wise analysis.

For tractability, a stratified subset of 60,000 samples was drawn and split 80/20 into training and hold-out test partitions. Preprocessing comprised removal of duplicates and missing records, standardisation of all features using training statistics only, and assignment of features to three growth-stage blocks based on acquisition dates.

Table: Class composition of the experimental subset

Class	Subset samples	Training	Test	Share of subset
Corn	11,400	9,120	2,280	19.0%
Peas	4,200	3,360	840	7.0%
Canola	10,800	8,640	2,160	18.0%
Soybeans	13,500	10,800	2,700	22.5%
Oats	4,500	3,600	900	7.5%
Wheat	10,200	8,160	2,040	17.0%
Broadleaf	5,400	4,320	1,080	9.0%

The above table illustrates that the subset preserves class proportions through stratification. It also exposes moderate imbalance, with minority classes such as peas and oats, which motivates the use of macro-averaged metrics.

Table: Distribution of features across growth-stage blocks

Stage	Window	Optical features	Radar features	Total d_s
Early	Vegetative	8	50	58
Mid	Flowering to peak canopy	20	42	62
Late	Maturity	18	37	55
Total	Full season	46	129	175
Mean per stage	Not applicable	15.3	43.0	58.3

The above table shows the candidate pool available to each stage-wise search. Radar variables dominate early stages, when canopy cover is limited, whereas optical variables become more numerous as the canopy develops.

Evaluation Metrics

Accuracy measures overall correctness, and macro-F1 averages class-wise F1 scores equally, which is essential under class imbalance. Cohen's kappa corrects for chance agreement. Subset size $|z^*|$ quantifies compactness and therefore sensing and computation cost. Hypervolume and

spread assess Pareto quality, with higher hypervolume indicating a better and broader front. The Explainability Consistency Index (ECI), as defined in the methodology, measures the stability of SHAP rankings across folds.

Baseline and Comparison Models

CAMO-FS is compared with seven configurations: the full 175-feature set, mutual-information filtering (top 40), recursive feature elimination (RFE), random-forest embedded importance, a single-objective genetic algorithm (GA), and a context-free global NSGA-II. All baselines use the same gradient-boosted classifier, so differences reflect the selection strategy rather than the learner.

Statistical Validation

Fitness estimates use stratified 5-fold cross-validation. Final results are averaged over ten independent runs with different random seeds, and 95% confidence intervals are obtained from 1,000 bootstrap resamples of the test set. Paired Wilcoxon signed-rank tests compare CAMO-FS with the strongest baseline, the context-free NSGA-II, and all reported improvements in accuracy and ECI reach $p < 0.01$.

Overall Predictive Performance

The lead-in below compares all selection strategies on the hold-out set.

Table: Overall performance of feature selection strategies on the test partition

Method	**Features retained**	**Accuracy (%)**	**Macro-F1**	**Kappa**	**ECI**
All features	175	96.8 ± 0.2	0.954	0.961	0.62
Mutual information	40	95.1 ± 0.3	0.941	0.942	0.58
RFE	52	96.0 ± 0.3	0.950	0.953	0.66
Random-forest embedded	48	96.2 ± 0.2	0.953	0.955	0.71
Single-objective GA	61	96.7 ± 0.2	0.954	0.960	0.69
Global NSGA-II	47	97.0 ± 0.2	0.957	0.964	0.78
CAMO-FS	42	97.6 ± 0.2	0.962	0.970	0.91

The above table shows that CAMO-FS attains the highest accuracy, macro-F1, kappa, and ECI while using only 42 of 175 features, about 24%. It even exceeds the full-feature model, indicating that removing redundant variables reduces noise rather than information.

Stage-Wise Selection Outcomes

The lead-in below reports what CAMO-FS retains at each growth stage.

Table: Stage-wise subset composition and stage-level accuracy

Stage	**Candidate d_s**	**Retained**	**Optical retained**	**Radar retained**	**Stage accuracy (%)**
Early	58	12	2	10	91.4
Mid	62	17	9	8	96.9
Late	55	13	6	7	95.8
CAMO-FS (all stages)	175	42	17	25	97.6
Global NSGA-II	175	47	14	33	97.0

The above table illustrates that the framework adapts selection to context. Early stages rely mainly on radar features, the mid stage balances optical and radar information, and the mid stage is the most discriminative single window. Compared with the global search, CAMO-FS retains more optical variables and fewer radar variables, reflecting its stage-aware balance.

Pareto Front Quality

The lead-in below evaluates the multi-objective search itself.

Table: Pareto-front quality and computational cost of search strategies

Method	**Hypervolume**	**Spread**	**Pareto set size**	**Knee macro-F1**	**Runtime (min)**
Weighted-sum GA	0.412	0.71	1	0.950	38
Global NSGA-II	0.584	0.52	23	0.957	61
Global NSGA-II with SHAP prior	0.611	0.49	26	0.959	66
Stage-wise NSGA-II without prior	0.629	0.46	31	0.960	74
CAMO-FS	0.673	0.41	34	0.962	71

The above table shows that CAMO-FS achieves the largest hypervolume and the lowest spread value, indicating a broader and more evenly distributed front. Both stage partitioning and SHAP priors contribute individually, and their combination gives the best front at a runtime comparable to the stage-wise search without a prior.

Class-Wise Performance

The lead-in below examines whether gains are uniform across crops.

Table: Class-wise F1 scores on the test partition

Class	**Test support**	**All features**	**RFE**	**Global NSGA-II**	**CAMO-FS**

Corn	2,280	0.975	0.972	0.977	0.981
Peas	840	0.932	0.928	0.939	0.945
Canola	2,160	0.973	0.970	0.975	0.978
Soybeans	2,700	0.979	0.977	0.981	0.983
Oats	900	0.915	0.910	0.921	0.929
Wheat	2,040	0.965	0.961	0.968	0.971
Broadleaf	1,080	0.937	0.933	0.941	0.947

The above table demonstrates that CAMO-FS improves every class. The largest gains appear in the minority classes, peas, oats, and broadleaf, which are the hardest to separate, so the improvement is not driven solely by majority classes.

Ablation Study

The lead-in below isolates the contribution of each component by removing it in turn.

Table: Ablation of CAMO-FS components

Variant	**Accuracy (%)**	**Macro-F1**	**ECI**	**Features retained**
Full CAMO-FS	97.6	0.962	0.91	42
Without stage partition	97.0	0.957	0.78	47
Without SHAP prior	97.2	0.960	0.86	45
Without stability objective f_3	97.5	0.961	0.72	40
Without knee selection (max-accuracy pick)	97.8	0.964	0.83	68

The above table shows that each component has a distinct role. Removing the stage partition causes the largest accuracy drop, removing f_3 mainly harms ECI, and replacing the knee rule with an accuracy-only pick marginally increases accuracy at the cost of 26 additional features and lower stability.

VII. Interpretation of Results

The results confirm that context-aware, multi-objective selection improves both efficiency and trustworthiness. CAMO-FS surpasses the full-feature model while retaining only 24% of the variables, implying that many acquired features are redundant or noisy. The stage-wise composition shows that radar information dominates early development, whereas the mid-season window carries the strongest combined signal, which is agronomically plausible. The ablation study attributes accuracy gains chiefly to stage partitioning and stability gains to the f_3 objective and SHAP prior. The Pareto analysis shows that the hybrid design produces a richer trade-off front without a prohibitive runtime. Gains on minority classes indicate improved robustness under imbalance. The ECI of 0.91 shows that the selected subsets yield reproducible explanations, which is valuable for practitioners.

VIII. Conclusion and Future Work

This paper presented CAMO-FS, a novel hybrid framework that combines growth-stage partitioning, SHAP-guided NSGA-II search, and knee-point selection to jointly optimise accuracy, compactness, and explanation stability. Evaluated on the UCI fused optical-radar crop mapping dataset, it retained 42 of 175 features, achieved 97.6% accuracy, a macro-F1 of 0.962, and an ECI of 0.91, outperforming all baselines with statistical significance. The ablation study confirmed that each component contributes measurably.

Future work will extend the framework to regression targets such as yield and stress indices, incorporate meteorological and soil variables, and validate transferability across regions and seasons. Adaptive stage detection from data rather than fixed windows, many-objective formulations that include sensing cost, and deployment on edge devices for real-time advisory are also promising directions.

A. Figures and Tables

TABLE I. Table Type Styles

Table Head	**Table Column Head**		
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[a.] Sample of a Table footnote. (*Table footnote*)

Fig. 1. Example of a figure caption. (*figure caption*)

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